DO NOW — RECALL

Lesson 145 · Packet Switching, Circuit Switching and Re

Name: _____ Class: _____ Date: _____

1. What is the job of DNS?

2. What is the accumulator for?

3. Which is faster in the worst case on sorted arrays, linear or binary search?

4. Which logic gate outputs 1 only when both inputs are 1?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Map names such as example.org to relevant resource records including IP addresses.

2. Model answer

Holding a CPU's working data/ALU results.

3. Model answer

Binary search: O(log n) rather than O(n).

4. Model answer

AND.

By the end, you can…

• Large message split into addressed packets for shared links.

• Packets share capacity and may arrive out of order.

• TCP can reorder and retransmit data, while IP itself is best-effort.

Vocabulary — know these terms

packet switching — Data divided into packets that may take different network routes.

circuit switching — A dedicated end-to-end circuit established before communication.

latency — Delay experienced by data in transit.

retransmission — Sending data again when loss is detected by a reliable protocol.

NEW LEARNING

Read, annotate and explain

Packetise data

Large message split into addressed packets for shared links.

Route over network

Packets share capacity and may arrive out of order.

Reassemble reliably

TCP can reorder and retransmit data, while IP itself is best-effort.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A 3-packet file crosses a packet-switched network. Packets arrive 2,1,3. Explain handling.

Send packets 1,2,3

Identify the relevant information and make a prediction.

Receive 2,1,3

Explain the change using correct technical terms.

Reorder 1,2,3

Packets can follow different routes or encounter different queue delays. With a reliable transport such as TCP, sequence numbers permit correct reordering before delivery of the byte stream; missing data can be retransmitted.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define packet switching and circuit switching, highlighting two differences. [6]

Question 2

Explain why packets in a message may reach the destination out of order. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate packet switching versus circuit switching for video calls, including reliability and resource efficiency. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

IP alone guarantees that every packet is received in the original order.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.